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Regulating Professional Players in Peer-to-Peer Markets: Evidence from Airbnb

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Abstract. We study professional players and their roles in peer-to-peer (P2P) markets. Most notably, P2P home-sharing platforms (e.g., Airbnb) consist of both professional hosts and nonprofessional individual hosts. What are the roles of the professionals? Should home-sharing platforms regulate their participation? Professional hosts may primarily offer properties that nonprofessional hosts would not supply and attract more guests—the differentiation effect. Or they may mostly supply similar properties and compete with the nonprofessionals—the competition effect. Using a unique data set of Airbnb listings, we first find that professional hosts’ properties are more expensive and have superior characteristics than nonprofessionals’. Second, we capitalize on a quasi-experiment in which Airbnb capped the number of properties a host can manage in several cities in the United States to determine the roles of professional hosts. With different predictions (about the policy impacts) under the differentiation versus competition effects, we find evidence suggesting the dominance of the latter. In particular, the policy increased the supply from nonprofessional hosts, and the price level of nonprofessional properties as a group went up after the policy. However, our findings of heterogeneity in policy impacts suggest that the dominance of competition is less prominent in certain markets. Finally, we find that the platform was not worse off in attracting reservations or securing revenue after the policy. Our findings contribute to both theory and practice as they reveal the roles of professional players and how P2P platforms can manage their participation.

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Keywords: peer-to-peer markets • professional players • competition • differentiation • platform economics • Airbnb

1. Introduction

Peer-to-peer (P2P) markets or the so-called “sharing economy,” such as Airbnb, Uber, and Prosper, provide platforms for grassroots individuals to share their underutilized resources for extra income (Einav et al. 2016). In recent years, the supply side of most P2P markets consists of not only individual, nonprofessional agents who own limited resources, but also professional players who typically operate multiple units of assets. These professional players arise to eradicate the original nature of P2P markets and increasingly defeat the grassroots force in the sharing economy (Athwal 2014, Li et al. 2019). For example, vacation rental investors (e.g., Evolve, Sonder, and Vacasa) account for a nonnegligible market share on

home-sharing platforms (Kieler 2017). Car rental companies (e.g., Avis and Hertz) allow individuals to use their vehicles directly on Uber’s platform (Schulz 2021). Institutional investors (e.g., Wells Fargo and Morgan Stanley) dominate P2P lending and equity crowdfunding (Athwal 2014).¹ The very institutions that P2P platforms aimed to disrupt seem now to prevail on these platforms.

What are the roles of professional players in a P2P market? Should platforms regulate professional players by restricting their participation? These are largely unanswered questions in the literature yet centric to the value proposition of sharing economy. In the realm of home sharing, on the one hand, professional hosts may offer properties that would not be provided

by nonprofessional hosts, for example, properties of higher quality or with superior features. We name this the “differentiation effect” through which professional players help attract guests who would not be attracted to home-sharing platforms if nonprofessional hosts were the only option. On the other hand, professional hosts may primarily provide similar properties and compete with their nonprofessional counterparts. With their professionalism, potential economies of scale (from managing multiple properties), and other advantages, professional hosts stand a higher chance of winning the competition and may eventually drive out mom-and-pops. We name this the “competition effect.”

Despite the distinctive relationships with nonprofessional hosts, professional hosts as a group may play both roles on home-sharing platforms, that is, differentiation and competition.² For example, a large professional host can manage a wide range of properties on a home-sharing platform (or across multiple sites) to target different travelers (Wong 2017). Some of these properties may have superior features with which no nonprofessional hosts can compete (i.e., differentiation), whereas others are more similar to the nonprofessionals’ and, hence, compete with them directly (i.e., competition). However, it is unclear which effect dominates and whether professional hosts can indeed play both roles on home-sharing platforms.

Practically speaking, understanding the roles of professional hosts in home sharing is crucial for platforms and society. Home sharing is shown to impact the economy (Barron et al. 2021, Chen et al. 2022) and has a wide variety of externalities (Filippas and Horton 2017), oftentimes leading to legal settlements with cities and regulators (Carville et al. 2020). Therefore, how to regulate home-sharing platforms, the vendors in particular (i.e., professional and nonprofessional hosts), is widely debated (Hui et al. 2016, Han and Zhang 2020, Sokol and Van Alstyne 2020). If the paradigm of home sharing involves a shift toward services being provided more and more by professional hosts, understanding their impact in the market and what platform intervention may improve the welfare of market participants is crucial. For example, if competition dominates the role of professional hosts, restricting their participation may help individual, nonprofessional hosts to survive on the platforms. If differentiation is the main role, in contrast, it would hurt the platforms and travelers (i.e., the consumers) by regulating professional hosts.

In this paper, we study the roles of professional hosts by leveraging a unique policy on the largest home-sharing platform: Airbnb. As of October 2020, Airbnb provides access to more than seven million properties in more than 100,000 cities in 220 countries and regions and is widely regarded as the unicorn of

the sharing economy (Isaacson 2017). Professional hosts operating multiple listings are a major driver of Airbnb’s growth. Yet Airbnb is facing unique challenges as these hosts continue to proliferate. City legislators in the United States have accused Airbnb of enabling the multiproperty practice, allowing a group of profit-seekers (i.e., professional hosts) to earn millions while cutting down housing inventories available to local residents.³ To soften the tension with local governments, in April 2016, Airbnb announced a policy that would cap the number of properties a host can manage in New York City and San Francisco to one (at one address). This so-called “one host, one home” policy (the OHOH policy or the policy hereafter) was later implemented in these two cities in November 2016 and, subsequently, in Portland, Oregon in February 2017.

Our empirical strategy examines the policy effects on various market outcomes to determine the roles of professional hosts. Specifically, the two distinctive effects, differentiation and competition, lead to different predictions about the policy impacts. For example, if the competition effect dominates, the OHOH policy encourages nonprofessional hosts to participate by restricting the supply from their professional competitors. As a result, the policy holds unchanged or even increases the total supply on Airbnb (aggregating professional and nonprofessional supply) if the increase in nonprofessional supply compensates for the decrease in professional supply. Conversely, if the differentiation effect dominates, the supply from nonprofessional hosts does not necessarily change after the policy because they operate in a different segment (and, thus, do not compete directly with professional hosts). Consequently, the total supply likely decreases because of the restricted supply from professional hosts. In other words, economically meaningful impacts of the policy (on supply and other market outcomes) serve as evidence of whether the major impact of professional players is differentiation or competition.

To examine the policy impacts, our research design is a quasi-experimental method that leverages the policy. Unlike the strategy studying the entry of a market player (which is popular in the literature, e.g., Greenwood and Wattal 2017, Zervas et al. 2017, Burtch et al. 2018), our empirical opportunity—the OHOH policy that restricts the participation of professional hosts—mitigates challenges such as simultaneity and endogenous agent decisions to enter a market (Berry 1992). In our research design, we construct a comparison group of markets that can mirror those in the policy-affected cities but have not been subject to the same restriction. Our main specification is a difference-in-differences (DID) method using a propensity score matching (PSM)-generated sample. We deploy several alternative specifications, for example, a relative time model, to verify our findings.

Our data come from multiple sources. We first obtain a proprietary Airbnb data set containing property-level performance metrics that are commonly used in the hospitality industry (e.g., nightly rate or price, number of reservation days, and revenue), property and host characteristics, and the geolocation of each property. To account for local factors, we complement the Airbnb data set with a rich set of economic and demographic characteristics collected from the American Community Survey (ACS) hosted by the U.S. Census Bureau. The final sample from combining the two data sets consists of more than 300,000 Airbnb properties in 534 zip codes from 10 cities (three policy-affected cities and seven unaffected cities) across the United States and spans a period from October 2014 to July 2017—a total of 34 months, covering a sufficiently large time window for the policy impacts to manifest.

Our empirical findings are mainly threefold. We first compare professional hosts on Airbnb with the nonprofessional cohort in terms of performance, property characteristics, and service features. We define professional hosts as those who ever operated more than one entire-home property in a city in any month during our study period (no need to manage multiple properties in all months).⁴ Based on the classification, about 17.54% of the hosts in our sample are labeled as professional hosts, who managed roughly 41.64% of the properties ever listed during our study period. We find that the properties managed by professional hosts have, on average, more bedrooms and more bathrooms and are more likely to be hotel-like (e.g., business-ready and instantly bookable) than those operated by nonprofessional hosts. Professional hosts are also more likely to be “super hosts” and respond more frequently to guest inquiries. They charge higher prices for their properties. Although their properties are more expensive, professional hosts receive more reservations and, eventually, make higher revenue than nonprofessional hosts. Overall, similar to what is documented in the literature (see, e.g., Li et al. 2019), we find that professional hosts outperform nonprofessional hosts by offering higher quality properties and achieving better performance.

From our main empirical analysis, the second set of findings concerns the policy’s impacts on multiple market outcomes (to determine the roles of professional hosts). The premise of all policy impacts is that the policy is effective in restricting the participation of professional hosts. Note that the policy caps the number of properties a host can manage (to one property at one address) but does not exclude or ban professional hosts on the platform. Not surprisingly, we find that the policy led to a significant reduction in the supply from professional hosts, evaluated using the number of properties they manage in a market. Interestingly,

the number of professional host accounts did not change after the policy, confirming that the policy is indeed not a ban on professional hosts.

More importantly, our findings of the policy impacts on supply and various price levels are consistent with the predictions under the dominance of the competition effect. The first evidence is that the OHOH policy increased the supply from nonprofessional hosts and the total supply. Similarly, we find that the median price of nonprofessional properties increased after the policy, which contradicts the prediction under the dominance of the differentiation effect (that the policy has no impact on the price level of nonprofessional hosts’ properties). We continue to find that the policy increased the median price of all properties, whereas the price level of professional host properties did not change after the policy. Finally, we find that the median price of top nonprofessional host properties increased more than that of bottom nonprofessional host properties,⁵ indicating that professional hosts may primarily compete with nonprofessional hosts who manage more expensive properties.⁶

Yet the competition effect is not the whole picture. Arguably, in markets with more heterogeneous properties (with a broader range of prices or other characteristics), the trade-off between differentiation and competition should favor the former. To test this prediction, we construct a price dispersion measure as a proxy for the heterogeneity of properties in a market. In a market with higher price dispersion, we note that the difference between the price distribution of professional properties and that of nonprofessional properties is arguably larger, which indicates less direct competition, or more differentiation, between the professional and nonprofessional hosts. Consistent with our prediction, using the price dispersion measure as a moderator, we find that the increases in supply and various price levels (of nonprofessional and all properties) are all smaller or even insignificant in markets with higher price dispersion. These findings suggest that the dominance of the competition effect is indeed less prominent or even muted in more heterogeneous markets.

What are the implications for home-sharing platforms by understanding the role of professional hosts? Our third set of analyses seeks answers to this question. For platforms, if the loss of supply from professional hosts can be compensated by the supply from nonprofessional hosts (because of the dominance of the competition effect), regulating professional hosts by implementing policies such as the OHOH may not reduce total reservations or overall revenue. Indeed, we find that the total reservations or revenue in a market did not change significantly after the policy. Furthermore, we find that the platform could be worse

off in more heterogeneous markets by restricting professional hosts' participation (because the dominance of competition is less prominent).⁷

Our study makes important contributions to both theory and practice. First, to the best of our knowledge, we are the first study on the roles of professional players in a P2P market. Guided by the theoretical framework (which formalizes the competition and differentiation roles), our empirical findings suggest that competition dominates the impact of professional hosts on Airbnb, which implies that mom-and-pops may be driven out by the professionals. These findings have implications for platform governance. Several other P2P markets also take action to support their nonprofessional vendors. In marketplace lending, for example, Prosper (the first such platform in the United States) has taken steps to cap the fraction of institutional investment in a personal loan and aims to win back retail investors. Our findings further inform that regulating professional players, in a way similar to the OHOH policy, may not hurt the overall revenue of the platform because nonprofessional players may serve the unmet needs of the consumers. Second, our paper suggests supplier management needs to be strategic. Our findings of heterogeneous policy impacts (in particular, that the platform may get hurt in more heterogeneous markets) suggest that P2P platforms can launch programs to promote differentiation. In more heterogeneous markets, the platforms may adopt policies to encourage professional players' participation as they offer services with differentiated characteristics to satisfy consumer needs. Airbnb, for example, launched "Airbnb Luxe," which targets high-end guests using professional property offerings. As professional players continue to rise, we consider our study as a call for future studies on further understanding the dynamics among professional vendors, nonprofessional vendors, and P2P platforms.

The rest of the paper is organized as follows. We begin with summaries of the related literature and highlight our contributions in the next section. Section 3 introduces the empirical setting (with a focus on the OHOH policy) and the data. Section 4 provides a framework to understand the roles of professional hosts. We discuss our empirical strategies and present the associated findings in Section 5. Section 6 concludes and discusses the implications of our empirical findings. Finally, we organize additional tests supporting the main results in the online appendix.

2. Related Literature

This paper is mainly related to three streams of literature. Our work first relates to the literature on supplier behavior in multisided markets as Airbnb exemplifies a two-sided market. Much of this literature

models the behavior of suppliers (firms or individuals), including multihoming (Landsman and Streimersch 2011), cross-side externalities (Rochet and Tirole 2003, Armstrong 2006, Rysman 2009), and responses to entry (Dafny 2005; Simon 2005; Hamilton and McManus 2005; McCann and Vroom 2010; Ellison and Ellison 2011; Seamans 2012, 2013, Prince and Simon 2015). Limited research empirically examines heterogeneous suppliers in multisided markets. One exception is Li et al. (2019), who study the difference in pricing between professional and nonprofessional hosts on Airbnb. We differ from their study by examining different questions that speak to the roles of professional players in home sharing, which manifests in not only pricing, but also other important market outcomes, such as supply, reservations, and revenue. Our unique empirical setting (i.e., the OHOH policy) allows us to demystify the underlying mechanisms driving the market equilibrium changes and the associated implications for regulating professional players.

Our paper also contributes to the emerging literature on P2P markets (Einav et al. 2016). This literature views P2P platforms as providing technologies that facilitate individual suppliers to capitalize on their underutilized resources at sufficiently low costs (Athey and Gans 2010, Seamans and Zhu 2014). Existing research focuses primarily on the competition between the disruptive P2P platforms and incumbent businesses that offer similar products or services (see, e.g., Zervas et al. 2017, Li and Srinivasan 2019) and the externality to local markets (see, e.g., Filippas and Horton 2017, Chen et al. 2022). Our study adds to the existing research by shifting the focus from the impact on the external environment (incumbents, local markets, etc.) to the internal platform.

Finally, P2P markets are stark examples of platforms that provide marketplaces for two sides of the market to transact without traditional intermediaries. With this view in mind, our study contributes to the extensive literature on platforms by studying a less understood aspect: platform governance and, particularly, the management of suppliers on a platform. The traditional view on the interaction between a platform and its suppliers centers on the competition between the two parties, for example, the platform's decisions to compete with their suppliers and the consequences of their entry into complementary markets (Yoffie and Kwak 2006, Pierce 2009, Cennamo and Panico 2015, Edelman and Lai 2016, Li and Agarwal 2017, Wan and Wu 2017, Zhu and Liu 2018, Parker and Van Alstyne 2018, Cennamo and Santalo 2019). However, research investigating platforms as regulators to manage their suppliers and intervene in the market is relatively scant (Carbó-Valverde et al. 2009), primarily because of the lack of empirical opportunities. Against the backdrop of the looming regulation on platform-based markets

(MacCarthy 2019), we utilize an empirical setting in which a P2P platform (Airbnb) purposefully regulates its suppliers to smooth the relationship with legislators (i.e., the OHOH policy). By uncovering the consequences of the platform-mandated regulation, our results have important implications for similar platforms.

3. Institutional Background

3.1. The One Host, One Home Policy

Airbnb has changed the way people travel by allowing them to “live like a local” (Benner 2016). Some hosts capitalize on the burgeoning of home sharing by renting out multiple properties on the platform. In the early days, these professional hosts played a significant role in helping Airbnb grow and scale. Whereas much of Airbnb’s business is based on nonprofessional hosts offering a spare room in their homes, these highly profitable outliers (i.e., professional hosts) have become the subject of controversy (Grothaus 2015). In particular, their practices of shifting housing inventory from local residential markets to traveler-oriented Airbnb draw extensive criticism from housing affordability advocates and legislators (Ting 2016). These professional hosts are particularly accused of resulting in a housing shortage and, hence, worsening housing affordability (Kerr 2017). As a result of these accusations, Airbnb has engaged in a series of lobbying campaigns or court battles against the cities (see examples in Carville et al. 2020).

To smooth the relationship with legislators who are concerned about the housing affordability challenges, Airbnb initiated a platform technique between 2016 and 2017 to restrict hosts from listing multiple properties at different addresses, that is, the so-called one host, one home policy.⁸ That is, the maximum number of addresses listed under each host ID was capped at one.⁹ Under this policy, a professional host who is willing to downsize to only one property does not need to exit the platform. Violating the policy may lead to a host’s properties being removed and Airbnb account being suspended or deactivated. The policy was first announced on April 2, 2016, and then implemented on November 1, 2016, in New York City and San Francisco (Airbnb 2016, 2017). On January 30, 2017, this policy was implemented and became immediately effective in Portland (Blanchard 2017). There were no other platform policies like one host, one home in these three cities between 2016 and 2017.

3.2. Data and Measures

We collected data on Airbnb listings through AirDNA, a research company that provides data and analytics services about short-term rental markets worldwide. AirDNA’s data service is widely used and endorsed by leading industry clients in real estate and hospitality, including Blackstone, Bank of America,

Merrill Lynch, Los Angeles Tourism & Convention Board, VACASA, and CBRE Group. Our data are at the individual property level and contain proprietary performance information, for example, monthly prices, transaction volume (the number of reservation days),¹⁰ and the revenue of each property. We also observe, for each property, detailed property characteristics (e.g., the number of bedrooms and bathrooms; maximum number of guests allowed; security deposit; minimum nights of stay required; number of photos displayed; whether a property is business-ready; whether a property is instantly bookable; and room type as an entire home, a shared room, or a private room), host service characteristics (e.g., whether a host is a super host and the response rate), and the unique geo-coordinates of a property. We follow Li and Srinivasan (2019) to define active listings as the properties that are marked as either booked or available for booking on the host’s calendar.

We complement the AirDNA data set with a rich set of variables from the ACS hosted by the U.S. Census Bureau. The ACS data set keeps track of annual economic and demographic characteristics of a zip code (e.g., population; number of households; median household income; unemployment rate; home vacancy rate; and fractions of White, Hispanic, and female population). We control these variables in our estimations because the hospitality industry (including Airbnb) is not immune to the economic conditions and demographic characteristics of local markets.

We match these two data sources at the zip code by year-month level: the unit of analysis in our main empirical tests. Our observation period is between October 2014 and July 2017 (a total of 34 months), which covers 18 months before and 16 months after April 2016 (the announcement of the policy). We choose this relatively large time window to observe the full impact of the policy. During our study period, there were altogether 308,328 Airbnb properties managed by 205,629 hosts (on average, roughly 1.5 properties per host) in 534 zip codes of the sampled cities, including 266 zip codes in three affected cities (New York City, San Francisco, and Portland) and 268 zip codes in seven unaffected cities. For easy reference, we define our variables in Table 1. We take logarithms of several variables with skewed distributions and denote them using “(logged)” throughout the paper.

4. Professional Hosts on Airbnb

We formally define professional hosts as those who ever managed more than one entire home in a city in any month during our study period. If a host managed two entire-home properties in San Francisco in January 2016, we label the host a professional host (and the properties as professional host properties)

Table 1. Variable Definitions

Variable	Definition
The policy	
<i>1(Policy)</i>	A dummy variable equal to one for zip codes in New York City or San Francisco after April 2016 and in Portland after February 2017
Market outcomes	
<i># properties (logged)</i>	The logarithm of the number of properties
<i># hosts (logged)</i>	The logarithm of the number of hosts
<i>Nightly rate (logged)</i>	The logarithm of the median nightly rate
<i># of reservation days (logged)</i>	The logarithm of the number of reservation days
<i>Total revenue (logged)</i>	The logarithm of the total revenue
Economic and demographic characteristics (ACS controls)	
<i>Population (logged)</i>	The logarithm of the population
<i># Households (logged)</i>	The logarithm of the number of households
<i>Median family income (logged)</i>	The logarithm of the median household income
<i>Unemployment rate</i>	Percentage of the labor force that is jobless
<i>House vacancy rate</i>	Percentage of the homeowner inventory that is vacant for sale
<i>% White population</i>	Percentage of the White population
<i>% Hispanic population</i>	Percentage of the Hispanic population
<i>% Female</i>	Percentage of the female population
Airbnb property characteristics	
<i># Bedrooms</i>	The number of bedrooms
<i># Bathrooms</i>	The number of bathrooms
<i>Max. guests</i>	Maximum number of guested allowed to stay
<i>Security deposits</i>	Deposit of money required
<i>Cleaning fee</i>	Expenses acquired to clean a property after a guest checks out
<i>Extra people fee</i>	Expenses acquired to add one more guest
<i>Minimum stay</i>	The number of nights of stay required
<i># Photos</i>	The number of property photos displayed online
<i>Business ready status</i>	Whether the property meets standards and amenities for business travelers
<i>Instant book enabled</i>	Whether the property requires no host approval before it can be booked
<i>Super host status</i>	Whether a host can provide extraordinary experiences for guests
<i>Response rate %</i>	The number of inquiries a host has responded to within 24 hours divided by the total number of inquiries a host has received in the past 90 days.

throughout the entire study period between October 2014 and July 2017. However, we do not label the host as a professional host if the host never managed two properties or more simultaneously in a month.

Our definition is the result of several considerations that aim to facilitate estimations of the policy impacts (to determine the roles of professional hosts). First, because only hosts who managed multiple properties *simultaneously* on Airbnb were subject to the policy restriction, our definition does not label those who operated different properties in different months (but never operated them simultaneously) as professional hosts. Second, we exclude most of the hosts who manage multiple listings at the same address (e.g., multiple rooms belonging to one property under the same address but listed separately on Airbnb) by requiring the property type to be *entire homes*, which is consistent with the policy target. Finally, if a host managed two or more entire-home properties at different addresses

in any month during the study period, the host is labeled as professional throughout our sample. The purpose is to keep the label consistent.¹¹

4.1. Comparing Professional with Nonprofessional Hosts

Table 2 compares professional with nonprofessional hosts. Professional hosts indeed exhibit a higher level of “professionalism.” First, professional hosts’ properties are of higher quality and in better conditions. They have more bedrooms and bathrooms, can accommodate more guests, are more likely to be hotel-like (i.e., business-ready and instantly bookable), and provide more photos. They tend to set up stricter rules of stay (i.e., higher security deposit, longer minimum nights of stay). Second, professional hosts provide better services. They are more likely to be super hosts and respond more often to guest inquiries. Third, potentially because professional hosts provide better services

and their properties are of higher quality, they charge higher prices than nonprofessional hosts (with a significant discrepancy of about \$64 per night). Despite their properties being more expensive on average, professional hosts receive more reservations in a month (evaluated using the number of reservation days). Hence, a property managed by a professional host generates higher revenue.

4.2. The Roles of Professional Hosts

The comparisons in the previous section showcase the differences between professional and nonprofessional hosts. In this section, we develop a theoretical framework on two potential roles of professional hosts in the form of their relationship with nonprofessional hosts. These discussions generate predictions on how the policy impacts various market outcomes, which we test in the following sections.

4.2.1. Differentiation vs. Competition. First, professional hosts may primarily supply properties and provide services that nonprofessional hosts cannot offer. For example, some vacation rental investors and companies list high-quality properties to differentiate from nonprofessional hosts and attract consumers who are willing to pay a premium. As the summary statistics in Table 2 suggest, the average nightly rate of professional hosts’ properties is higher than that of nonprofessional

host properties. If the primary role of professional hosts is to offer properties that nonprofessional hosts do not supply, these two cohorts do not compete with each other directly. We name this the differentiation effect. This effect implies that professional hosts play the role of expanding the home-sharing market to consumers who would not be attracted to the market if the properties of nonprofessional hosts were the only option.

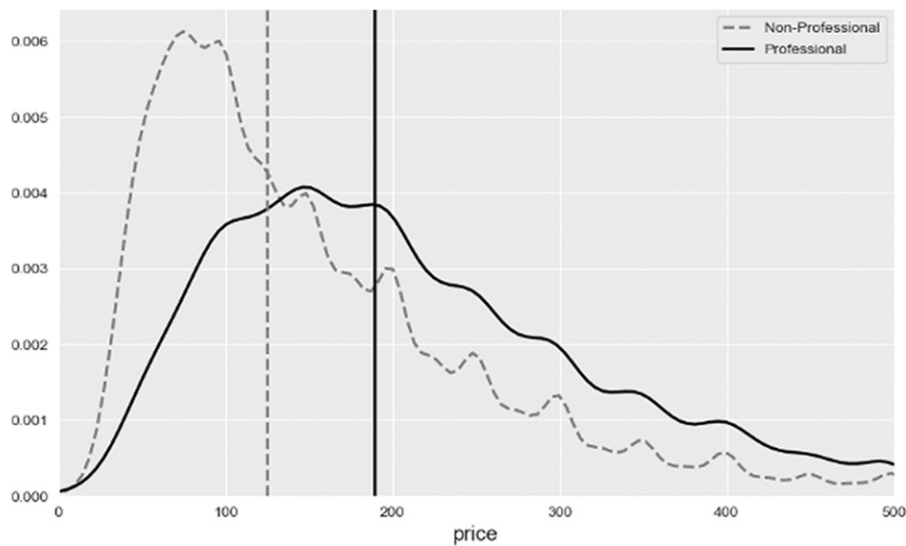
Conversely, if professional hosts mainly provide properties and services that nonprofessional hosts can also offer, these two cohorts compete for similar guests in a home-sharing market. In this case, professional hosts arguably stand a higher chance of winning the competition because of their competitive advantages. For example, professional hosts typically manage multiple properties simultaneously, which may grant them better economies of scale and more experience of property operations than nonprofessional hosts. They may also gain margins over nonprofessional hosts from their expertise in the relevant industry and dedication to property management in the home-sharing market. Regardless of the specific advantages, we name this the competition effect of professional hosts.¹² A consequence is that the presence of professional hosts may eventually crowd out nonprofessional hosts.

Figure 1 compares the price distributions (nightly rates) of properties managed by professional hosts with those

Table 2. Comparing Properties Managed by Professional Hosts with Properties Managed by Nonprofessional Hosts

	Properties by professional hosts				Properties by nonprofessional hosts				<i>t</i> -stat	<i>p</i> -value
	Mean	Standard deviation	Minimum	Maximum	Mean	Standard deviation	Minimum	Maximum		
Property performance										
<i>Nightly rate</i>	268.996	426.744	1	11,000	204.603	372.269	0	12,000	23.758	<2.220e-16
<i># Reservation days</i>	8.660	8.892	0	31	6.329	7.974	0	31	41.098	<2.220e-16
<i>Revenue</i>	1,826.280	2,971.173	0	180,000	918.405	1,977.886	0	250,000	49.712	<2.220e-16
Property characteristics										
<i># Bedrooms</i>	1.496	1.101	0	10	1.197	0.736	0	14	44.065	<2.220e-16
<i># Bathrooms</i>	1.356	0.761	0	8	1.209	0.546	0	15.5	31.119	<2.220e-16
<i>Max. guests</i>	4.224	2.656	1	18	2.827	1.832	1	16	85.271	<2.220e-16
<i>Security deposits</i>	445.293	598.225	95	10,000	382.037	493.064	25	5,100	12.672	<2.220e-16
<i>Cleaning fee</i>	91.445	69.447	5	1,800	57.960	48.318	5	1,318	69.885	<2.220e-16
<i>Extra people fee</i>	28.330	24.816	3	300	28.491	26.482	1	300	−0.634	0.526
<i>Minimum stay</i>	3.981	10.771	0	540	3.099	10.971	0	1,250	12.534	<2.220e-16
<i># Photos</i>	16.830	13.917	0	255	10.687	9.716	0	231	69.320	<2.220e-16
<i>Business ready status</i>	0.187	0.390	0	1	0.092	0.289	0	1	39.366	<2.220e-16
<i>Instant book enabled</i>	0.195	0.397	0	1	0.098	0.298	0	1	39.404	<2.220e-16
Service characteristics										
<i>Super host status</i>	0.163	0.369	0	1	0.093	0.291	0	1	30.025	<2.220e-16
<i>Response rate %</i>	91.912	18.074	0	100	87.963	23.856	0	100	27.785	<2.220e-16
<i>Number of observations</i>	29,228				124,236					

Notes. Each observation is the median value of the corresponding variable up until the policy announcement (October 2014–March 2016). For example, the variable *Nightly rate* is the median nightly rate (price) of a property across all months between October 2014 (the beginning of our study period) and March 2016 (the month right before the policy announcement). The sample we use in this table contains hosts who were active before the announcement of the policy. The purpose is to eliminate any contamination from the policy.

Figure 1. Price Distributions of Professional and Nonprofessional Host Properties

by nonprofessional hosts. The two vertical lines are the median nightly rates across the respective cohorts of hosts. Clearly, the distribution of professional host properties is located to the right of the distribution of nonprofessional hosts, indicating that professional host properties are overall more expensive. The median nightly rate of professional host properties is also greater than that of nonprofessional hosts.¹³

The difference between the price distributions (of professional and nonprofessional properties) in Figure 1 seems to suggest that the primary role of professional hosts is differentiation because their properties are overall more expensive. However, this could also be the outcome of competition. As we argue previously, professional hosts may crowd out nonprofessional hosts because of their competitive advantages. A consequence is that nonprofessional hosts are forced to focus on the lower end of the market as the figure shows. Regardless, one cannot determine the roles of professional hosts by comparing the characteristics or performance metrics (supply, prices, etc.) because of the observational nature of those data. The OHOH policy, our research context, offers a unique chance to uncover the relationship between professional and nonprofessional hosts by restricting the participation of one party but not the other.

4.2.2. Predictions About the Policy Impacts on the Supply. In this section and the next, we develop a series of predictions about the policy impacts on various market outcomes (mainly supply and price levels), which we test in Section 5. First, the OHOH policy's impact on the supply from professional hosts is clear. Because, under the policy, a host can manage at most one property (at one address) in a city, the professionals have to cut down on the number of listings they

manage on Airbnb. Naturally, the policy decreases the supply from professional hosts—the premise of other policy effects.

Second, regarding the policy impacts on the supply of nonprofessional hosts, the dominance of differentiation indicates that professional and nonprofessional hosts mainly operate in different segments of the market. In this scenario, by restricting the operations of one party (professional hosts), the policy does not impact the other party's (nonprofessional hosts) decisions to participate. In other words, if we find that the supply from nonprofessional hosts does not change significantly after the policy, the evidence may suggest the dominance of the differentiation effect.¹⁴ However, if the competition effect dominates, the policy encourages nonprofessional hosts to operate on the platform, causing an increase in their supply.

Furthermore, the policy's impact on total supply also differs under differentiation and competition. If the competition effect dominates, the increased supply from nonprofessional hosts and the decreased supply from professional hosts dictate an indeterminate prediction of the policy impact on total supply. The sign depends on the relative magnitudes. However, if the primary role of professional hosts is differentiation, the total supply likely decreases after the policy. This is because the supply from nonprofessional hosts remains largely unchanged under differentiation (as argued), whereas the supply from professional hosts likely decreases (again, the policy restricts their supply). Because the total supply is the sum of the supply from professional and nonprofessional hosts, if the policy decreases the total supply on the platform, the evidence suggests the dominance of differentiation. Conversely, if we fail to find the policy reducing the total supply,

the competition effect is likely the dominant role. As a summary, we predict the following.

Proposition 1A. *If differentiation dominates the role of professional hosts, on average across markets, the OHOH policy does not affect the supply from nonprofessional hosts in a significant manner. The policy decreases the total supply.*

Proposition 1B. *If competition dominates the role of professional hosts, on average across markets, the OHOH policy increases the supply from nonprofessional hosts. The impact on total supply is indeterminate, depending on the relative magnitudes of increased nonprofessional supply and decreased professional supply.*

4.2.3. Predictions About the Policy Impacts on Price Levels. We next reason how the policy impacts on various price levels may differ across the two different roles of professional hosts. The dominance of differentiation suggests that professional hosts primarily operate in a segment separate from nonprofessional hosts. Therefore, the policy has little or no impact on nonprofessional hosts (not only supply, but also prices). If we find that the policy does not have any significant impact on the price level of nonprofessional host properties, the evidence supports the dominance of differentiation.

On the contrary, if the competition role dominates, the direction of the policy impact on the price level of nonprofessional hosts depends on which group (of nonprofessional hosts) are the main competitors of professional hosts. If the nonprofessionals who manage relatively more expensive properties are the main competitors (of professional hosts), the policy encourages this subgroup of nonprofessional hosts to participate more. As a result, there is a larger fraction of more expensive nonprofessional properties, and the policy increases the price level of nonprofessional hosts as a whole group. Similarly, the policy decreases the price level of nonprofessional properties if those who manage relatively cheaper properties are the main competitors of professional hosts. Our evidence so far suggests that professional properties are, on average, more expensive than nonprofessional properties (Table 2, Figure 1). One reasonably expects that the nonprofessionals who manage relatively more expensive properties are the main competitors of the professionals (under the domination of competition). Hence, we predict that the OHOH policy likely increases the price level of nonprofessional properties.

Regarding the impact on the price level of the entire market, because professional hosts' properties are, on average, more expensive than those of nonprofessional hosts (Table 2), we expect that the policy likely decreases the overall price level if differentiation dominates. The reason is that, with fewer expensive

properties (from professional hosts) and a largely stable supply of less expensive properties (from nonprofessional hosts), the market has a larger fraction of cheaper properties after the policy; as a result, the price level of the entire market drops. However, if we find the price level of the market increases after the policy, the evidence, again, contradicts the prediction under the dominance of the differentiation effect and, hence, suggests that the competition effect is likely to prevail. Therefore, we predict the following.

Proposition 2A. *If differentiation dominates the role of professional hosts, on average across markets, the OHOH policy does not affect the price level of nonprofessional host properties and decreases the price level of all properties.*

Proposition 2B. *If competition dominates the role of professional hosts, on average across markets, the OHOH policy increases the price level of nonprofessional host properties. The price level of all properties also increases after the policy.*

5. Results and Findings

As a road map of our empirical analyses, before testing the predictions developed in the previous section, we begin by discussing the main specification of our empirical model in Section 5.1. Section 5.2 reports our check on the premise of the policy impacts: whether it restricts the supply from professional hosts. In Section 5.3, we formally test the theoretical predictions and report the policy's impacts on supply and various price levels.¹⁵ We then explore the trade-off between differentiation and competition by studying heterogeneity in the policy impacts in Section 5.4. Section 5.5 reports findings concerning managerial implications. We rule out alternative explanations in Section 5.6.1 and present a series of robustness checks in Section 5.6.2.

5.1. The Main Empirical Design

The variation in policy announcement or implementation across markets in different periods offers a quasi-experimental opportunity to examine the roles of professional hosts. Our empirical strategy is the standard DID approach capitalizing on the policy. The policy was only effective in New York City, San Francisco, and Portland, that is, the policy-affected cities. We first rank cities in the United States by Airbnb market size (the total number of Airbnb properties) and economic and demographic characteristics, including population; number of households; median household income; unemployment rate; home vacancy rate; and fractions of the White, Hispanic, and female population. Then, we use a Euclidean distance algorithm to find the "nearest neighbor" cities that can best mirror the three cities experiencing the policy. This national search yields an unaffected group of seven major cities,

Table 3. Comparability Between Affected and Unaffected Zip Code Groups

	211 affected zip codes		211 unaffected zip codes		Mean difference	<i>p</i> -value
	Mean	Standard deviation	Mean	Standard deviation		
Total properties (logged)	3.776	1.657	3.733	1.388	0.043	0.775
Population (logged)	10.389	0.804	10.286	0.759	0.103	0.174
# Households (logged)	9.426	0.745	9.323	0.683	0.103	0.140
Median family income (logged)	10.978	0.436	10.956	0.471	0.022	0.613
Unemployment rate	9.062	3.479	9.295	4.534	−0.233	0.555
House vacancy rate	8.201	3.908	8.336	4.782	−0.135	0.751
% White population	36.418	23.855	37.342	24.905	−0.924	0.697
% Hispanic population	18.162	14.891	18.147	18.013	0.015	0.993
% Female	51.274	2.891	50.887	3.023	0.387	0.179

including Los Angeles; San Diego; San Jose; Seattle; Washington, D.C.; Boston; and Philadelphia.

To construct the main sample for analysis, we use a propensity score matching algorithm to further strengthen comparability (Abadie and Imbens 2006). The matching is based on observed local market characteristics, including the Airbnb market size and economic and demographic factors, the same set of variables used in the national search of unaffected cities.¹⁶ The propensity scores are predicted probabilities of being subject to the policy based on a logistic regression. The outcome is a set of markets (zip codes) in affected and unaffected cities that are most comparable in those observed local characteristics.

The PSM-generated sample contains 211 zip codes in the three affected cities and 211 zip codes in unaffected cities (i.e., a total of 422 zip codes in the matched sample). The matched sample covers almost 80% ($\approx 422/534$) of all zip codes in the original sample. Table 3 compares the affected and unaffected groups based on the mean values of all observed characteristics before April 2016, when the policy was first announced. After matching, we can see that the covariates are quite balanced.¹⁷ The two groups of zip codes are not significantly different in terms of the Airbnb market size or economic and demographic characteristics.¹⁸

Using the PSM sample, the main specification of our empirical model is a DID regression. Our unit of analysis is at the zip code by year-month level. The equation is as follows:

$$Y_{it} = \mu_i + \nu_t + \beta \cdot 1(\text{Policy})_{it} + \gamma' \mathbf{X}_{it} + \varepsilon_{it}. \quad (1)$$

In the equation, Y_{it} is the dependent variable. Depending on the specific regression, we examine the policy's impacts on various market outcomes (e.g., supply, various price levels, number of reservation days, and revenue). We include zip code fixed effects, μ_i , and year-by-month fixed effects, ν_t , as in standard DID designs. The policy dummy, $1(\text{Policy})_{it}$, is the key regressor, and its estimated coefficient captures the

policy impact on the specific market outcome (dependent variable). Note that the policy dummy is zip code-by-month specific: it takes the value of one for zip codes in New York City and San Francisco after April 2016 and zip codes in Portland after February 2017; it equals zero elsewhere, including the zip codes of unaffected cities in all periods, the observations for the zip codes in New York City and San Francisco before April 2016, and the zip codes in Portland before February 2017.¹⁹

In the set of control variables ($\mathbf{X}_{it}$), we first include city-specific time trends. The purpose is to control for any unobserved trends that are different from city to city (see, e.g., Zervas et al. 2017, Chen et al. 2022). Operationally, the city-specific time trends are constructed by interacting a set of 10 dummy variables, each indicating a city, with a variable of sequential numbering of the months that begins with one for the first month in our study period, that is, October 2014. We also include the set of ACS variables, as defined in Table 1, to control for local economic and demographic factors. The residual term, ε_{it} , captures idiosyncratic shocks.

5.2. The Premise of Policy Impacts

The premise of any policy impacts is that the policy effectively restricts the participation of professional hosts. If the policy is not effective, the supply from professional hosts in policy-affected markets should stay constant relative to unaffected markets. So we test whether the supply of professional hosts, in both sheer numbers and fractions, dropped after the policy.

Our main measures of supply are the number of properties and the number of host accounts. Table 4 reports the regression results based on our main specification as Equation (1). Column (1) reports the results from the regression of the (logged) number of properties by professional hosts (in a zip code in a month) on the regressors. We find the coefficient for the policy dummy negative and statistically significant, indicating that the policy is effective in restricting the supply

Table 4. Policy Effects on the Supply of Professional Hosts

Dependent variables: (all logged)	Supply of professional hosts	
	(1) # Properties	(2) # Hosts
$1(\text{Policy})_{it}$	−0.0623** (0.0279)	−0.0189 (0.0211)
ACS controls: Local socioeconomic and demographic characteristics		
Population (logged)	0.7655 (0.6065)	1.4912*** (0.4956)
# Households (logged)	0.6678 (0.6772)	−0.8417* (0.4839)
Median family income (logged)	−0.2102 (0.2795)	−0.4679** (0.2160)
Unemployment rate	−0.0247* (0.0138)	−0.0238** (0.0102)
House vacancy rate	0.0081 (0.0102)	0.0072 (0.0070)
% White population	0.0154* (0.0091)	0.0209*** (0.0072)
% Hispanic population	0.0174 (0.0129)	0.0114 (0.0100)
% Female	−0.0037 (0.0121)	−0.0055 (0.0090)
Zip code fixed effects	Yes	Yes
Year-by-month fixed effects	Yes	Yes
City time trends	Yes	Yes
Observations	14,163	14,163
Adjusted R ²	0.9549	0.9682

Notes. Robust standard errors clustered at the zip code level are shown in parentheses. All continuous variables are winsorized at 1% and 99% levels.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

from professional hosts. Numerically, the properties of professional hosts decreases by about 6.23% in a zip code in a month after the policy. Column (2) presents the results from the regression of the (logged) number of professional host accounts on all regressors. As we argue in Section 3, the policy restricts but does not ban professional hosts from operation. The insignificant coefficient of the policy dummy supports our argument, showing no systematic exits of professional hosts after the policy. Taken together, the results confirm the effectiveness of the policy.

To verify the robustness of the finding, we use the fraction of professional hosts (in both property count and host accounts) as alternative dependent variables. The results are reported in Online Table A3. The estimated effects of the policy remain qualitatively unchanged (i.e., the decreased fractions of professional hosts), further confirming the effectiveness of the policy.²⁰

5.3. Evidence on the Roles of Professional Hosts

5.3.1. Evidence from the Policy Impacts on Supply.

Our main analysis explores the policy effects on multiple market outcomes to determine the roles of professional hosts in home-sharing markets. Following the predictions in Section 4.2.2, we start with the policy's

impact on the supply of nonprofessional hosts to test which effect dominates. Panel (I) in Table 5 reports the regression results. Similar to Table 4, we adopt two measures for the supply from nonprofessional hosts: the number of properties and the number of host accounts in a zip code in a month. Columns (1) and (2) present the results from the regression of these two measures on the regressors (Equation (1)), respectively. The positive coefficients of the policy dummy in both regressions suggest that the policy increases the supply from nonprofessional hosts, lending support to the dominance of the competition effect (consistent with Proposition 1B). Numerically, the number of nonprofessional host properties increases by about 4.31%, and the number of nonprofessional host accounts increases by 5.5%.

Panel (II) of Table 5 reports the regression results of the policy impacts on total supply. Again, we use the total number of properties and the total number of host accounts (in a zip code in a month) as two measures. Interestingly, we find that, although the total number of properties does not change significantly, the increase in the total number of host accounts is statistically significant (more than a 4% increase in the policy-affected markets relative to unaffected markets).²¹ Because none of the two policy effects on total

Table 5. Policy Impacts on the Supply

Dependent variables: (all logged)	Panel (I): Supply of nonprofessional hosts		Panel (II): Total supply	
	(1) # <i>Properties</i>	(2) # <i>Hosts</i>	(3) # <i>Properties</i>	(4) # <i>Hosts</i>
$1(\text{Policy})_{it}$	0.0431* (0.0222)	0.0550** (0.0213)	0.0131 (0.0200)	0.0410** (0.0194)
ACS controls: Local socioeconomic and demographic characteristics				
<i>Population (logged)</i>	−0.0859 (0.5358)	−0.1782 (0.4971)	−0.0200 (0.5177)	0.0439 (0.4811)
# <i>Households (logged)</i>	−0.3493 (0.6315)	−0.2511 (0.6013)	0.0500 (0.6104)	−0.3863 (0.5680)
<i>Median family income (logged)</i>	−0.5108** (0.2197)	−0.4973** (0.1987)	−0.5245** (0.2275)	−0.5806*** (0.1959)
<i>Unemployment rate</i>	−0.0348*** (0.0100)	−0.0286*** (0.0085)	−0.0326*** (0.0095)	−0.0300*** (0.0083)
<i>House vacancy rate</i>	−0.0109 (0.0093)	−0.0132 (0.0086)	−0.0042 (0.0090)	−0.0094 (0.0079)
% <i>White population</i>	−0.0032 (0.0080)	−0.0024 (0.0076)	−0.0021 (0.0080)	−0.0009 (0.0076)
% <i>Hispanic population</i>	0.0307*** (0.0099)	0.0295*** (0.0093)	0.0312*** (0.0098)	0.0285*** (0.0091)
% <i>Female</i>	−0.0147 (0.0116)	−0.0139 (0.0106)	−0.0143 (0.0111)	−0.0142 (0.0102)
Zip code fixed effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
City time trends	Yes	Yes	Yes	Yes
Observations	14,163	14,163	14,163	14,163
Adjusted R^2	0.9667	0.9714	0.9723	0.9755

Notes. Robust standard errors clustered at the zip code level are shown in parentheses. All continuous variables are winsorized at 1% and 99% levels.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

supply is negative, the evidence lends further support to the dominance of the competition effect (Proposition 1B).

5.3.2. Evidence from the Policy Impacts on Price Levels.

As with supply, the two roles—differentiation and competition—generate different predictions on the policy impacts on various price levels as discussed in Section 4.2.3. Column (1) in Table 6 reports the regression results with the dependent variable being the price level of nonprofessional host properties. The specific measure is the (logged) median nightly rate among all nonprofessional host properties in a zip code in a month.

We note that the coefficient of the policy dummy is positive and statistically significant. As we argue, the economically meaningful change in the price level of nonprofessional host properties rejects the prediction under the dominance of differentiation (Proposition 2A), offering another piece of evidence consistent with the dominance of competition (Proposition 2B). Quantitatively, the policy increases the median price of nonprofessional host properties by more than 7.3%.

We also regress the market level price on the policy dummy and other regressors as in Equation (1) and report the results in column (3) of Table 6. The dependent variable is the median nightly rate of all properties (including professional and nonprofessional) in a zip code in a month. The positive coefficient of the policy dummy suggests that the policy increases the market-level price by almost 6% (statistically significant). Again, the evidence is consistent with our other findings and further supports the dominance of competition (Proposition 2B).

An implicit assumption of our argument (that the increased market-level price contradicts the predictions under the dominance of differentiation) is that the price level of professional host properties stays relatively constant despite the policy. If the assumption is violated, for example, professional hosts keep their most expensive properties under the policy restriction (thus, increasing the median price of professional hosts' properties), then the increased market-level price can be attributed to either the competition effect or the increased price level of professional host properties. We test this assumption by examining the policy's impact on the price level of professional host

Table 6. Policy Effects on the Price Levels

	(1) Nonprofessional host properties	(2) Professional host properties	(3) All properties
Dependent variables: <i>Median nightly rate (logged)</i>			
$1(\text{Policy})_{it}$	0.0737*** (0.0178)	−0.0090 (0.0861)	0.0592*** (0.0181)
ACS controls: Local socioeconomic and demographic characteristics			
<i>Population (logged)</i>	0.0888 (0.3409)	1.1921 (1.6642)	0.0768 (0.3537)
<i># Households (logged)</i>	0.1496 (0.3746)	−1.2254 (1.4457)	0.1827 (0.3698)
<i>Median family income (logged)</i>	0.3241*** (0.1195)	−0.3789 (0.6280)	0.3363*** (0.1219)
<i>Unemployment rate</i>	0.0059 (0.0068)	−0.0834** (0.0394)	0.0084 (0.0071)
<i>House vacancy rate</i>	−0.0027 (0.0045)	0.0261 (0.0226)	0.0013 (0.0044)
<i>% White population</i>	−0.0008 (0.0059)	−0.0109 (0.0284)	0.0019 (0.0059)
<i>% Hispanic population</i>	−0.0054 (0.0059)	−0.0021 (0.0299)	−0.0061 (0.0058)
<i>% Female</i>	0.0009 (0.0051)	0.0208 (0.0289)	0.0042 (0.0055)
Zip code fixed effects	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes
City time trends	Yes	Yes	Yes
Observations	14,163	14,163	14,163
Adjusted R ²	0.7339	0.6798	0.7686

Notes. Robust standard errors clustered at the zip code level are shown in parentheses. All continuous variables are winsorized at 1% and 99% levels.
* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

properties. Similarly, we calculate the median nightly rate of professional host properties in a zip code in a month as the dependent variable. As column (2) of Table 6 shows, we fail to find a significant impact of the policy on the price level of professional host properties. The evidence is consistent with the assumption and, therefore, confirms the dominance of competition.

5.3.3. Who Competes with Professional Hosts? Our empirical findings consistently suggest the dominance of the competition effect, meaning that professional hosts primarily provide properties that nonprofessional hosts may also supply so that the two cohorts potentially compete for similar guests. A natural question is who competes with the professional hosts. Are the nonprofessional hosts as a whole group or a subgroup of them the main competitors of professional hosts? Summary statistics suggest that professional host properties are overall more expensive (Table 2, Figure 1). One may reasonably expect that professional hosts compete primarily with those nonprofessional hosts who provide properties at a price level closer to theirs (i.e., relatively more expensive properties). Similarly, we find that the policy increases the price level of nonprofessional host properties (column (1) of Table 6). The evidence suggests that the policy encourages the nonprofessional hosts operating more

expensive properties to participate more in the market, which also implies the competition between this subgroup of nonprofessional and professional hosts.

We test the conjecture (that professional hosts primarily compete with the subgroup of nonprofessional hosts operating more expensive properties) by studying whether there are differential policy impacts on nonprofessional hosts offering more expensive properties versus those offering less expensive properties. Specifically, we separate nonprofessional host properties (in each zip code in each month) into two subgroups: one group with nightly rates higher than the median nightly rate of all nonprofessional host properties (labeled “top”) and the other group lower than the median (labeled “bottom”). Therefore, for each zip code, we have exactly 50% top and 50% bottom nonprofessional host properties in each month.

If professional hosts primarily compete with top nonprofessional hosts, we expect that the policy increases the price level of top more than bottom nonprofessional hosts (as our arguments for Proposition 2B).²² We adopt similar price measures as before, namely, the (logged) median nightly rate of top nonprofessional host properties in a zip code in a month and the same rate for bottom nonprofessional host properties. Panel (I) in Table 7 presents the results from two separate regressions on the two subgroups. Consistent with

Table 7. Policy Effects on the Prices and Reservations of Top and Bottom Nonprofessional Hosts

Dependent variables: (all logged)	Panel (I): Price of nonprofessional host properties		Panel (II): Reservation of nonprofessional host properties	
	(1) Top	(2) Bottom	(3) Top	(4) Bottom
$1(\text{Policy})_{it}$	0.1014*** (0.0230)	0.0673** (0.0340)	0.1191*** (0.0441)	0.0123 (0.0491)
ACS controls: Local socioeconomic and demographic characteristics				
Population (logged)	0.1636 (0.4761)	−1.2562 (0.8518)	−0.1611 (0.8511)	−0.0020 (0.8726)
# Households (logged)	0.0145 (0.5215)	0.5214 (0.8388)	0.6392 (0.9878)	−0.1355 (0.9025)
Median family income (logged)	0.2527 (0.1595)	−0.5772* (0.3472)	0.6513* (0.3352)	−0.1659 (0.3757)
Unemployment rate	0.0080 (0.0095)	−0.0025 (0.0193)	0.0021 (0.0178)	0.0178 (0.0167)
House vacancy rate	−0.0076 (0.0061)	−0.0005 (0.0132)	0.0055 (0.0126)	0.0079 (0.0122)
% White population	−0.0068 (0.0089)	−0.0427*** (0.0164)	0.0104 (0.0139)	−0.0121 (0.0118)
% Hispanic population	−0.0074 (0.0084)	0.0045 (0.0166)	−0.0108 (0.0165)	−0.0090 (0.0150)
% Female	−0.0119* (0.0070)	−0.0273 (0.0173)	0.0110 (0.0175)	−0.0032 (0.0165)
Zip code fixed effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
City time trends	Yes	Yes	Yes	Yes
Observations	14,163	14,163	14,163	14,163
Adjusted R ²	0.6656	0.4991	0.5410	0.4877

Notes. Robust standard errors clustered at the zip code level are shown in parentheses. All continuous variables are winsorized at 1% and 99% levels.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

our conjecture, we find that the policy has a greater impact on the price level of top (column (1)) than bottom nonprofessional hosts (column (2)). Numerically, after the policy, the price of top nonprofessional host properties increases by more than 10%, whereas the price of bottom nonprofessional host properties only increases by about 6.73%.

If professional hosts indeed mainly compete with top nonprofessional hosts, the more direct evidence should be that the “lost” demand for professional host properties, because of the policy restriction, is captured by top nonprofessional hosts. In contrast, the reservations of bottom nonprofessional host properties should not change significantly after the policy. We test these conjectures by examining the policy impacts on reservations (more specifically, the (logged) median number of reservation days) of the two subgroups of nonprofessional hosts. Panel (II) of Table 7 reports the regression results. Highly consistent with the predictions, we find that the median reservation days of top nonprofessional host properties surges by more than 11% (as in column (3)). In sharp contrast, the median reservation days of bottom nonprofessional host properties does not change significantly after the policy (as in column (4)). These results provide strong

evidence of the competition between professional and top nonprofessional hosts.

5.4. Is Competition the Whole Picture?

Is competition the whole story of professional hosts? Can the trade-off between differentiation and competition favor the former in some markets? These are natural questions one asks. In a market with more heterogeneous properties, the dominance of competition may be less prominent. To see this, first, because professional hosts’ properties are, on average, more expensive and superior to nonprofessional hosts’ properties (Table 2, Figure 1), the differences between their properties may be larger in more heterogeneous markets.

Then, the larger differences (in prices, quality, etc.) indicate that professional hosts are less likely to compete with nonprofessional hosts in such markets. In other words, the competition effect is likely less prominent or even muted in a market with more heterogeneous properties. We show that the OHOH policy increases the supply and price levels of nonprofessional hosts and those of all hosts. If the increases (in supply and price levels) are smaller or even insignificant in

more heterogeneous markets, the evidence lends support to our conjectures.

We construct a price dispersion measure to proxy for the heterogeneity in Airbnb properties in a market. The reason we focus on prices rather than other metrics is that prices are generally believed to be a useful information aggregator assembling different aspects of a product or service (Salop and Stiglitz 1977, Sandholm 2015). Specifically, for each zip code, we calculate the standard deviation of the price distribution across properties listed before April 2016 (when the policy was announced) as the measure, named “dispersion.” A greater value of the price dispersion measure (i.e., a larger standard deviation of the price distribution) indicates the market having more heterogeneous properties and vice versa.²³ We find that our price dispersion measure negatively correlates with the p -value from two-sample Kolmogorov–Smirnov tests on the price distributions of professional and nonprofessional properties (with null hypotheses that the two distributions do not differ). This finding suggests that, as price dispersion in a zip code increases, the price distributions of professional and nonprofessional properties are less likely to be the same, which confirms that professional host properties are more different from nonprofessional host properties in a market with more heterogeneous properties.

With the price dispersion measure, $Dispersion_{it}$, we explore its moderating role in our main policy impacts. The corresponding regression equation is adapted from the main specification (Equation (1)) by interacting the policy dummy with the moderator:

$$Y_{it} = \mu_i + \nu_t + \beta_1 1(\text{Policy})_{it} + \beta_2 1(\text{Policy})_{it} \cdot Dispersion_{it} + \gamma' \mathbf{X}_{it} + \varepsilon_{it}. \quad (2)$$

Note that, because $Dispersion_{it}$ does not vary across different periods (i.e., each zip code has one value of this price dispersion measure), the stand-alone term of $Dispersion_{it}$ is not identified with zip code fixed effects and, thus, omitted. The coefficient of the interaction term, β_2 , captures the moderating role of the price dispersion measure in the policy effects on the corresponding market outcome, Y_{it} . In this set of analyses, we examine the same set of dependent variables as our main analysis (Tables 5 and 6). Under our conjecture that the trade-off between differentiation and competition favors the former in more heterogeneous markets, we expect that the moderating effect of the price dispersion measure takes the opposite sign as our main policy impacts.

Table 8 reports the results from the regressions of various supply measures, including the supply from nonprofessional hosts and the total supply, using the specification of Equation (2). Panel (I) presents the effects on the supply of nonprofessional hosts, and

Panel (II) is for the total supply. Across four regressions, we find that, consistent with our prediction, the coefficients of the interaction terms are all negative and statistically significant (opposite to our main policy effects). In addition, as Panel (II) shows, the total supply may decrease in markets with sufficiently high price dispersion after the policy. For example, at the value of the 75th percentile price dispersion, the overall effect of the policy in such a market is about -6.37% ($\approx 11.94\% - 0.07\% \times 261.5$, statistically significant according to an F test). This evidence suggests that the differentiation effect may dominate the role of professional hosts in extremely heterogeneous markets. An implication is that, in such markets, nonprofessional hosts may not fill the void from the restricted supply from professional hosts.

The moderation of the price dispersion measure in the policy impacts on prices lends further support to our conjectures. Table 9 presents the regression results. Column (1) is for the regression of the median nightly rate of nonprofessional host properties, and column (2) reports the results from the regression of the median nightly rate of all properties. Similar to the moderation results in Table 8, the coefficients of the interaction terms, once again, take the opposite sign as the main policy effects.

5.5. Managerial Implications

So far, our empirical evidence points to the dominance of competition in the role of professional hosts although the dominance is less prominent or even muted in more heterogeneous markets. What are the implications for the platform? Is Airbnb necessarily better or worse off by implementing the OHOH policy? We show evidence that the restricted supply from professional hosts is compensated by the increased supply of nonprofessional hosts. As a result, the platform may not be worse off by adopting a restriction on professional hosts such as the OHOH policy.

We formally test the prediction by examining the policy impacts on total reservations and total revenue. The specific measure of reservations is the total number of reservation days (aggregated over all properties) in a zip code in a month, and the measure for total revenue is the sum of all properties' monthly revenue (= nightly rate $\times$ reserved days). Panel (I) of Table 10 presents the regression results. Column (1) reports the policy effect on total reservations, and column (2) is for the total revenue. Consistent with our prediction, the policy does not increase or decrease the reservations or the revenue on the platform given neither of the two policy coefficients are statistically significant. The findings imply that the increased supply from nonprofessional hosts indeed picks up the lost demand of professional properties.

Table 8. Moderating Effects of Price Dispersion on the Supply

Dependent variables: (all logged)	Panel (I): Supply of nonprofessional hosts		Panel (II): Total supply	
	(1) # <i>Properties</i>	(2) # <i>Hosts</i>	(3) # <i>Properties</i>	(4) # <i>Hosts</i>
$1(\text{Policy})_{it}$	0.1350*** (0.0322)	0.1443*** (0.0314)	0.1194*** (0.0308)	0.1369*** (0.0299)
$1(\text{Policy})_{it} \times \text{Dispersion}_i$	−0.0006*** (0.0002)	−0.0006*** (0.0002)	−0.0007*** (0.0002)	−0.0006*** (0.0002)
ACS controls: Local socioeconomic and demographic characteristics				
<i>Population (logged)</i>	−0.0089 (0.5312)	−0.1034 (0.4929)	0.0690 (0.5102)	0.1242 (0.4757)
# <i>Households (logged)</i>	−0.2585 (0.6258)	−0.1629 (0.5964)	0.1550 (0.6022)	−0.2916 (0.5626)
<i>Median family income (logged)</i>	−0.4855** (0.2193)	−0.4728** (0.1985)	−0.4953** (0.2265)	−0.5542*** (0.1956)
<i>Unemployment rate</i>	−0.0328*** (0.0099)	−0.0267*** (0.0084)	−0.0303*** (0.0094)	−0.0280*** (0.0081)
<i>House vacancy rate</i>	−0.0104 (0.0094)	−0.0127 (0.0088)	−0.0036 (0.0090)	−0.0089 (0.0081)
% <i>White population</i>	−0.0038 (0.0081)	−0.0030 (0.0076)	−0.0028 (0.0080)	−0.0015 (0.0076)
% <i>Hispanic population</i>	0.0278*** (0.0098)	0.0266*** (0.0093)	0.0278*** (0.0097)	0.0254*** (0.0091)
% <i>Female</i>	−0.0131 (0.0118)	−0.0123 (0.0108)	−0.0124 (0.0112)	−0.0125 (0.0104)
Zip code fixed effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
City time trends	Yes	Yes	Yes	Yes
Observations	14,163	14,163	14,163	14,163
Adjusted R^2	0.9671	0.9717	0.9728	0.9759

Notes. Robust standard errors clustered at the zip code level are shown in parentheses. All continuous variables are winsorized at 1% and 99% levels. Because the price dispersion measure does not vary across different periods (i.e., each zip code has one value of the price dispersion measure), the stand-alone term of the price dispersion measure is not identified under zip code fixed effects and, thus, omitted in the regressions.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Furthermore, our heterogeneity analysis (Tables 8 and 9) demonstrates the dominance of competition being less prominent in more heterogeneous markets. An implication is that the lost demand of professional hosts may not be fully recovered by nonprofessional hosts in such markets, potentially resulting in fewer reservations and less revenue in extremely heterogeneous markets; that is, the platform may get hurt by the policy. We test this conjecture by examining the moderating role of our price dispersion measure in the policy impacts on total reservations and total revenue. Panel (II) of Table 10 reports the regression results based on Equation (2). The coefficients of the interaction term between the policy dummy and the price dispersion measure are negative and statistically significant (as in columns (3) and (4)).

5.6. Further Analyses

To further support the main analyses, we rule out alternative explanations of our empirical findings and report the results from a series of robustness checks.²⁴

5.6.1. Excluding Alternative Explanations.

5.6.1.1. Relisting Properties by Professional Hosts.

The policy restricts a host from listing multiple properties at different addresses under the host's account. One may wonder whether some professional hosts can relist their properties targeted by the policy back to Airbnb by signing up new accounts under friends' or families' names or using fake names. We argue that such a relisting behavior, if any, is unlikely to bias our results. First, it is not costless to manage multiple properties using fake names or friends' names, legally and financially. Before listing a property, Airbnb usually asks hosts to verify their identities using legal documents such as driver license numbers, taxpayer identification numbers, phone numbers, etc. Hosts are also financially and legally liable to short-term rental taxes, registration fees, rental insurance payments, and lawsuits (or disputes) if incidents occur during guest stays. These legal and financial consequences make it costly for professional hosts to "game" the policy using others' names or fake names.

Table 9. Moderating Effects of Price Dispersion on Prices

	(1) Nonprofessional host properties	(2) All properties
Dependent variables: <i>Median nightly rate (logged)</i>		
$1(\text{Policy})_{it}$	0.1152*** (0.0224)	0.1039*** (0.0247)
$1(\text{Policy})_{it} \times \text{Dispersion}_i$	−0.0003** (0.0001)	−0.0003* (0.0002)
ACS controls: Local socioeconomic and demographic characteristics		
<i>Population (logged)</i>	0.1236 (0.3394)	0.1142 (0.3557)
<i># Households (logged)</i>	0.1906 (0.3675)	0.2269 (0.3614)
<i>Median family income (logged)</i>	0.3355*** (0.1193)	0.3486*** (0.1221)
<i>Unemployment rate</i>	0.0068 (0.0067)	0.0094 (0.0069)
<i>House vacancy rate</i>	−0.0025 (0.0045)	0.0015 (0.0044)
<i>% White population</i>	−0.0011 (0.0058)	0.0016 (0.0058)
<i>% Hispanic population</i>	−0.0068 (0.0059)	−0.0076 (0.0060)
<i>% Female</i>	0.0016 (0.0050)	0.0050 (0.0055)
Zip code fixed effects	Yes	Yes
Year-by-month fixed effects	Yes	Yes
City time trends	Yes	Yes
Observations	14,163	14,163
Adjusted R^2	0.7348	0.7696

Notes. Robust standard errors clustered at the zip code level are shown in parentheses. All continuous variables are winsorized at 1% and 99% levels.
* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Second, one may argue that, compared with using fake or friends’ names, it is relatively easy to relist the targeted properties using family members’ names. We explore our Airbnb data set to identify whether such a relisting behavior is in a significant manner among professional hosts and whether this gaming behavior, if identified, may bias our findings. The data set allows us to do so. Specifically, we observe the unique geo-coordinates, that is, latitudes and longitudes, of all sampled properties. For example, a property in New York City has the geo-coordinates of 40.693741 and −73.932907. Note that the latitudes and longitudes are at a high precision level of six decimal places. With six decimal places in these geo-coordinates, if two properties have the same latitude and longitude, their distance should be no more than 0.11 meters apart.²⁵ The high precision level allows us to tell whether two properties are at different addresses to a reasonable degree.

Using the geo-coordinates, we look for properties located at the same address, that is, the same latitude and longitude, but managed by different host accounts before and after the policy in the three affected cities. We suspect that only these properties are “relisted” ones. Our exact matching of the latitudes and longitudes at the six decimal places identifies two such

properties, which is negligible out of 180,264 properties in the three policy-affected cities.

Using the geo-coordinates, we further look for “possibly relisted” properties. Specifically, for each professional property removed after the policy and given a range around its geolocation, we find nonprofessional properties that are newly registered after the policy. This way, we can identify possibly relisted properties at any given precision. Setting the highest precision (at six decimal points or 0.11 meters) generates the same set of nonprofessional properties as before. If we relax the proxy distance to 1.1 meters (or five decimal points), we can identify 18 possibly relisted properties or 0.01% of all affected properties. At a distance of 11.1 meters (or four decimal points), we find 1,634 possibly relisted properties or 0.9% of all affected properties. All these cases generate negligible sets of possibly relisted properties.

Even if we relax the distance to 44.4 meters (a fairly large distance to separate two residential properties), there are only 7,030 properties that are possibly relisted. First, some of these 7,030 properties may not necessarily be relisted. In addition, among the 7,030 properties, there are only 5,787 unique new properties (because of overlaps resulting from the very coarse precision of 44.4 meters). Finally, the average gap

Table 10. Policy Effects on Total Reservations and Revenue and the Moderation of Price Dispersion

Dependent variables: (All logged)	Panel (I): Policy effects		Panel (II): Moderating effects	
	(1) # Total Reservation Days	(2) Total Revenue (\$)	(3) # Total Reservation Days	(4) Total Revenue (\$)
$1(\text{Policy})_{it}$	0.0108 (0.0564)	0.0037 (0.0562)	0.1646** (0.0837)	0.1437* (0.0832)
$1(\text{Policy})_{it} \times \text{Dispersion}_i$			-0.0010*** (0.0004)	-0.0009** (0.0004)
ACS controls: Local socioeconomic and demographic characteristics				
Population (logged)	-0.7403 (1.7438)	-0.7196 (1.7146)	-0.6115 (1.7427)	-0.6024 (1.7152)
# Households (logged)	1.5599 (1.6319)	1.2678 (1.5938)	1.7119 (1.6313)	1.4061 (1.5946)
Median family income (logged)	-1.7482*** (0.6000)	-1.8305*** (0.5987)	-1.7060*** (0.5960)	-1.7920*** (0.5955)
Unemployment rate	-0.0518* (0.0300)	-0.0486* (0.0289)	-0.0485 (0.0300)	-0.0456 (0.0290)
House vacancy rate	0.0069 (0.0258)	0.0031 (0.0250)	0.0076 (0.0254)	0.0038 (0.0248)
% White population	-0.0504* (0.0257)	-0.0465* (0.0251)	-0.0513** (0.0255)	-0.0474* (0.0249)
% Hispanic population	0.0325 (0.0296)	0.0293 (0.0293)	0.0276 (0.0295)	0.0248 (0.0293)
% Female	-0.0327 (0.0274)	-0.0331 (0.0276)	-0.0300 (0.0273)	-0.0306 (0.0276)
Zip code fixed effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
City time trends	Yes	Yes	Yes	Yes
Observations	14,163	14,163	14,163	14,163
Adjusted R ²	0.8639	0.8653	0.8642	0.8655

Notes. Robust standard errors clustered at the zip code level are shown in parentheses. All continuous variables are winsorized at 1% and 99% levels.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

between the removal date and the possibly relisted date of these 7,030 properties is 12 months, casting doubts on whether the majority of these properties were relisted by professional hosts if their goal is to dodge the OHOH policy and make a profit.

Still, we replicate our main analysis by excluding all possibly relisted properties identified using 44.4 meters as the proxy distance (which generates the largest set of suspiciously relisted properties). The results are reported in Online Tables A4 and A5. The estimated policy impacts remain highly consistent with our main results (Tables 4–6). Replicating the tests with stricter matching on shorter distances (e.g., 11.1 or 1.11 meters), which yield fewer possibly relisted properties, also produces highly consistent results (omitted because of space constraints but available upon request). All the evidence shows that the potential gaming behavior of professional hosts, if any, would not bias our results.

5.6.1.2. New Entrants. One may worry that the OHOH policy, by restricting the participation of professional hosts, introduced a different set of hosts to

the platform. These new hosts and their properties may be different from the incumbents. As a result, the policy effects we find could be driven solely by the new hosts, which obfuscates our exploration of the roles of professional hosts.

To rule out this alternative explanation, we test (1) whether there are indeed systematically more (or fewer) entrants in affected markets relative to unaffected markets after the policy and (2) whether their prices are significantly different (from those in unaffected markets). We define a host being an entrant in a month if the host's account was created in that month. The classification of entrant properties, thus, follows. Similar to the main analyses, we calculate the number of properties by entrants and the number of entrant host accounts in a zip code in a month as two measures of their supply. The price level of entrant properties is defined as the median nightly rate among all entrant properties in a zip code in a month. The results are reported in Online Table A6. Columns (1) and (2) present the results on the policy effects on the supply of entrants, and column (3) focuses on their

price level. None of the policy dummy coefficients are statistically significant, suggesting that our main findings are unlikely driven by the entrants.

5.6.1.3. Strategic Price Increase by Nonprofessional Hosts. We find that the OHOH policy increases the price level (median nightly rate) of nonprofessional hosts. There are two potential explanations of this specific policy impact. One is that, as we show in Section 5.3.3, the increase is driven by the top nonprofessional hosts (because they are likely the main competitors of professional hosts and more of them participate after the policy). The other is that nonprofessional hosts, as a group, strategically raise their listing prices after the policy restricted the professional cohort. To test the likelihood of the second explanation, we obtain a property-by-month panel from AirDNA and check if individual properties in affected cities increase their listing prices (i.e., nightly rate) after the policy relative to properties in unaffected cities. The regression model is similar to Equation (1) with the unit of analysis at the property by year–month level instead.

Online Table A7 reports the findings. Interestingly, rather than raising their listing prices, the negative policy dummy in column (1) indicates that the policy decreased the average nightly rate (in a month) of nonprofessional host properties. Similarly, the average nightly rate of professional host properties and all properties both reduce after the policy (columns (2) and (3)). These findings, hence, do not support the second explanation (that hosts strategically price higher after the OHOH restriction). Furthermore, we find that the average nightly rate of top nonprofessional host properties decreases after the policy, whereas the average nightly rate of bottom nonprofessional host properties does not change (columns (4) and (5) of Online Table A6). This is likely because the OHOH policy encourages more top nonprofessional hosts to participate (as we show in our empirical analysis) and, therefore, intensifies the competition among this subgroup. Regardless, these property-level results exclude the second explanation and lend support to our main arguments.

5.6.1.4. Placebo Tests by Randomizing the “Treated” Groups. To further ensure that it is the OHOH policy driving the results rather than some other factors not captured by the policy, we conduct a placebo test in which we run counterfactual simulations using only unaffected zip codes. In each draw, we randomly select 50% of the zip codes in unaffected cities and treat them as if they were subject to the policy. We keep the other half set of unaffected zip codes and label them as being unaffected. Then, we replicate our main analyses to obtain the coefficients of the policy

dummy (using the announcement date). We repeat this process multiple times.

Each figure in Online Figures A1 and A2 depicts the distribution of the estimated coefficients for the policy dummy from simulations of 1,000 draws each. We conduct the simulations on the main dependent variables, including the supply of nonprofessional and professional hosts, the total supply, and various price measures. As we can see from the figures, in all cases, most of the mass (of a distribution) lies in a small interval centered around zero. This evidence suggests that most of the estimated coefficients are not significantly different from zero in a statistical sense and implies that the “counterfactual” policy constructed in the simulations does not affect any of the dependent variables. Therefore, the results from this placebo test suggest no significant effects in the absence of the policy and, thus, it is the policy driving our findings.

5.6.2. Robustness Checks and Additional Analyses.

5.6.2.1. Using a Relative Time Model. An assumption of a DID design is that the affected and unaffected groups have relatively parallel trends before the event: the OHOH policy in our case. We rely on a relative time model (Angrist and Pischke 2008) to check whether the parallel prepolicy trends exist between the affected and unaffected zip codes. The model is specified as

$$Y_{it} = \mu_i + \nu_t + \sum_{\tau=-T}^N \lambda_{\tau} D_{it,\tau} + \gamma' \mathbf{X}_{it} + \varepsilon_{it}, \quad (3)$$

where $D_{it,\tau}$ is a dummy variable indicating, for policy-affected zip codes, whether month t is the τ th before or after the policy announcement in April 2016 for New York City and San Francisco (and implementation in February 2017 for Portland). For example, $D_{it,0}$ takes the value of one for the policy month (announcement or implementation) for all zip codes in the three policy-affected cities and zero otherwise. Note that these dummy variables are zeros for all unaffected zip codes in all periods. The set of coefficients λ_{τ} identifies whether a similar prepolicy trend exists in the affected and unaffected zip codes and how the policy impacts evolve.

We estimate Equation (3) with $T = 7$ and $n = 12$ to inspect the trends from eight months before the policy and one year after the policy. We also include a dummy for all months that are eight or more months before the policy ($\tau = -7^{+}$) and another dummy for months that are 12 months or more after the policy ($\tau = 12^{+}$). Considering there might be an immediate impact of the policy, we use $\tau = -1$ (the month before the policy) as the baseline month. We estimate the model for both supply and price levels from professional and

nonprofessional hosts, validating the parallel trends assumption for our main results.

Online Table A8 exhibits the relative time results on supply. As shown in columns (1) and (2) of Online Table A8, the supply from nonprofessional hosts in affected and unaffected zip codes are relatively parallel to each other before the policy. None of the coefficients λ_τ ($\tau < 0$) for nonprofessional host supply is significantly different from zero. After the policy, the nonprofessional host supply increases slowly but steadily compared with the prepolicy trend. Columns (3) and (4) reveal a similar pattern in the supply from professional hosts (opposite in the signs). For the periods before the policy, the majority of the coefficients λ_τ ($\tau < 0$) are also insignificant (with only two exceptions out of 14 coefficients in total), validating the parallel trends. After the policy, there is a sudden decline in professional host supply, confirming the premise of the policy. Finally, columns (5) and (6) report the results on total supply. Similarly, we find the parallel trends with 11 out of 14 prepolicy dummies statistically insignificant. After the policy, like our main findings in Table 5, the number of properties does not change significantly, whereas the number of hosts increases steadily over time. All these results suggest that there are unlikely prepolicy trends (different between affected and unaffected zip codes) that confound our policy impacts.

Online Table A9 reports the relative time results on prices. We also find that the prepolicy trends are overall parallel between affected and unaffected zip codes with almost all prepolicy coefficients (19 out of 21 in total) not statistically different from zero. After the policy, all time-dummy coefficients λ_τ ($\tau \geq 0$) are significantly greater than zero in the regression of nonprofessional host prices. In sharp contrast, none of the coefficients in the professional host price regression is significantly different from zero after the policy. Also, consistent with our main findings, we find that most of the coefficients after the policy are positive and statistically significant in the regression of the market-level price. These results further reveal the evolution of the outcomes after the policy, qualitatively consistent with our main findings (Tables 4–6).

5.6.2.2. DID Estimations on the Full Sample. The sample used in our main analysis (reported in Section 5) is a sample generated from propensity score matching between zip codes in policy-affected and unaffected cities. To ensure that the results are not biased by the chosen matching method, we reestimate the main policy effects using the full sample without the matching process. Specifically, we replicate the regressions of various supply and price measures on the policy dummy and other covariates as in Equation (1). Online Table A10 presents the results on the supply of

professional hosts, the supply of nonprofessional hosts, and the total supply. They are highly consistent with our main findings. Online Table A11 presents the policy effects on the median price of professional host properties, the median price of nonprofessional host properties, and the market-level price. Once again, we find qualitatively consistent results with our main findings.

5.6.2.3. DID Estimations on an Alternative PSM-based Sample.

We include a rich set of covariates in the propensity score matching process to generate our main sample (Table 2). One may suggest including more covariates for a potentially more precise matching between two groups of zip codes. Adding more matching covariates is, however, a double-edged sword: it allows us to find markets that are more comparable between two groups, but having more matching covariates shrinks the sample size dramatically (thus, potentially compromising the statistical power and trustworthiness of the results). Despite this consideration, we include an additional set of matching covariates, including the supply, prices, and reservations of nonprofessional hosts, professional hosts, and at the market level in the PSM process. The result is a sample containing 154 zip codes from affected cities and 154 zip codes from unaffected cities. Online Table A12 lists and compares all matching covariates between the two zip code groups for a balance check. Using this new matched sample, we further replicate the main regressions in Online Tables A13 and A14 and find largely consistent results with the findings using our main PSM sample.

5.6.2.4. Excluding San Francisco. We control for unobserved trends in Airbnb markets that are different from city to city using city-specific time trends (see, e.g., Equation (1)). One may still worry about the uniqueness of some affected cities. For example, one may argue that the Airbnb market of San Francisco is different from other cities in our sample because San Francisco is the hub of technology companies, and the hosts there are more “tech-savvy” than hosts in other cities. To alleviate the concern, we estimate the policy effects by excluding San Francisco from the sample. As a result, the remaining sample contains New York City, Portland, and the seven unaffected cities. We first rerun the propensity score matching to construct the sample without the zip codes in San Francisco and then replicate the main regressions (i.e., Tables 4–6). Online Tables A15 and A16 report the regression results for our policy effects on supply and prices, respectively. Not surprisingly, the results are highly consistent with our main findings, suggesting that including zip codes in San Francisco does not bias our results.

5.6.2.5. Excluding Hosts with Multiple Entire-Home Listings from the Same Address. A concern regarding our professional host labeling is that, without observing the exact address of each property, there might be hosts who manage multiple entire-home properties at the same address (e.g., multiple apartments in the same building sharing the same street address), which may not be subject to the policy. To address this concern, we first examine how many such cases exist in our data. Similar to the relisting check in Section 5.6.1, we use the geo-coordinates of properties in our sample to identify such listings. Using the six and five decimal places from our geolocation data, we find zero and two entire-home properties potentially using the same address, respectively. Using four decimal places to match addresses, we find 302 entire-home properties suspiciously using the same address. Then, we replicate the main analysis by excluding these 302 properties. The results are reported in Online Tables A17 and A18. The estimated policy impacts remain highly consistent with those reported in our main results (Tables 4–6). Thus, it is unlikely that the potential mislabels of these professional hosts, if any, bias our results.

5.6.2.6. An Alternative Definition of Professional Hosts. In our main analysis, we label a host a professional host if the host ever managed more than one entire home in a city in any given month during our study period. One may worry that the policy effects we discover are solely driven by the labeling. To ensure the robustness of our findings and alleviate the sample selection concern, we replicate our main analysis under a different definition of professional hosts. In the alternative definition, a host is designated professional if the host ever managed more than one listing in a city in a month during our study period. The difference from our main definition is that this alternative label does not have a requirement on the type of listings, which could be entire homes (as in the main definition), private rooms, or shared rooms.

We first repeat the comparisons between professional and nonprofessional host properties under the alternative label of professional hosts. Online Table A19 reports the summary statistics. Because the alternative label is less restrictive, the differences in the means of all variables between professional host properties and nonprofessional host properties are expected to be smaller than those reported in Table 2. This is because professional hosts labeled in Table 2 are, in essence, a subset of those labeled in Online Table A19. Under our main definition, the average monthly revenue of professional host properties is \$1,826.28, and that of nonprofessional host properties is \$918.41 with a difference of more than \$900 (as in Table 2). Indeed, under the alternative definition, the average monthly

revenue of professional host properties is \$1,265.87, which is only about \$300 higher than that of nonprofessional host properties (\$967.70).

We further repeat the main analysis (i.e., estimations of the policy effects on supply and prices in Tables 4–6) under the new definition of professional hosts. Online Table A20 presents the results on supply. As expected, with a less stringent definition of professional hosts, we find that the decrease in the supply by professional hosts is smaller than the estimate in our main definition. Meanwhile, we find that the policy effect on the supply of nonprofessional hosts remains qualitatively the same as our main findings. Note that the results on the market-level supply remain the same because they are not affected by our labeling of professional hosts. Furthermore, Online Table A21 reports the results on prices. Overall, we find that the impact on different price levels remains qualitatively the same as our estimation in the main specification.

5.6.2.7. Policy Effects on Prices Controlling the Total Supply. In our analysis of the policy effects on prices (as in Section 5.3), one may argue that the prices are a market equilibrium outcome, and controlling the total supply in the market alleviates the concern that the policy dummy may pick up some supply effect. Despite the simultaneity concern (i.e., demand and supply determine prices), we add the total supply, measured by the total number of properties (professional and nonprofessional) in a zip code in a month, as an extra control variable in the regressions for prices. Online Table A22 reports the results. We find highly consistent results with our main findings. The coefficients of the policy dummy in all three regressions are qualitatively the same as our main results reported in Table 6.

5.6.2.8. Estimations with Extra Controls of Airbnb Market Characteristics. Some may wonder whether our main findings of the policy effects still hold if extra Airbnb market characteristics are added in the estimation. The argument is that the market outcomes, such as supply and prices, are associated with the characteristics of the local Airbnb market. To alleviate the concern, we aggregate the property-level characteristics to the market level and add them as extra controls in our regressions. For example, we compute the average number of bedrooms and bathrooms of all properties in a zip code in a month as a local market characteristic. Similarly, we calculate other local market characteristics, such as the average security deposits, fraction of business-ready properties, and fraction of super host properties. Online Tables A23 and A24 report the regressions of supply and prices (as in our main analyses Tables 4–6). All the results continue to

support our main conclusion, that is, the dominance of competition in the role of professional hosts.

5.6.2.9. Policy Effects on Additional Market-level Outcomes. In the main results (Tables 4–6), we examine the policy's impact primarily on the supply and various price levels to detect the roles of professional hosts. Some may wonder about the policy effects on other market-level outcomes, such as the median reservations and the median revenue (rather than the "total" reservations and revenue examined in Table 10). Online Tables A25 and A26 report the results. The specific measures of reservations are the median reservation days of professional host properties in a zip code in a month and a similar quantity for nonprofessional host properties and all properties. Similarly, the revenue measures are the median revenue of professional host properties and that of nonprofessional host properties and all properties.

Interestingly, results reported in column (2) of Online Table A25 suggest that the policy increases the median reservation days of professional host properties, whereas the median reservation days of nonprofessional host properties stays relatively constant after the policy (as in column (1)). As a result, the median number of reservation days among all properties increases after the policy (as in column (3)). Note that this finding does not contradict the insignificant change in the total number of reservation days (aggregating reservation days of all properties) of Table 9 because the policy does not increase the total number of properties (as in Table 5). We further find that the median revenue of either professional or nonprofessional host properties did not change after the policy (as in Online Table A26). The median revenue of all properties also remains unchanged after the policy, consistent with our main finding that the policy does not affect the total revenue (adding up the monthly revenue of all properties). These findings also suggest that the OHOH policy is primarily a shock to the supply side of the market and does not affect the demand side in a significant manner.

6. Conclusions

As the sharing economy proliferates, platforms need to understand the new internal environment in which professional players rise to challenge the nature of P2P sharing. Our research is geared precisely toward aligning supplier management strategies to this new environment. Using a unique policy shock (capping the participation of professional hosts), we empirically investigate the roles of professional hosts on Airbnb, the largest P2P home-sharing platform.

Our findings first unveil the superiority of professional hosts to their nonprofessional counterparts in

terms of property condition, service quality, and performances (e.g., more reservation days and higher revenue). We then theorize and empirically test whether the superiority indicates the dominance of a competition or differentiation role of professional hosts. To answer this question, we investigate the impacts of Airbnb's one host, one home policy on primarily two sets of market outcomes: supply and various price levels. We find that, although the policy restricts the supply from professional hosts, it encourages the supply from nonprofessional hosts, which compensates for the restricted supply from professional hosts. As a result, the total supply in the market does not decrease. Simultaneously, the policy drives up the price level of nonprofessional host properties and the market-level price. Particularly, professional hosts seem to compete mainly with top nonprofessional hosts who offer more expensive properties. All the evidence points consistently to the dominance of competition in the role of professional hosts. Yet the competition role is not the whole picture. In markets with highly heterogeneous properties, the trade-off between differentiation and competition may favor the former. Finally, despite the policy precluding some superior suppliers (the professionals) from the market, the platform performance (in terms of reservations and revenue) remains relatively stable after the policy. This is potentially because the loss of policy-affected professional hosts was well-captured by the rise of nonprofessional hosts after the policy. However, because differentiation can dominate the role of professional hosts in extremely heterogeneous markets, the platform is hurt in attracting reservations and securing revenue in those markets.

As P2P markets continue to grow, how to manage suppliers who are largely unregulated in the sharing economy represents a critical issue for the platforms. Our research, being the first comprehensive study of how regulating professional players impacts P2P market outcomes, provides new insights into the challenge faced by the platforms. First, our empirical findings suggest that, for Airbnb, regulating professional hosts encourages nonprofessional hosts to participate, which is simultaneously revenue-secured (overall for the platform). Therefore, there is potentially a need for home-sharing platforms to manage the participation of professional hosts. That said, a careful interpretation of our results in terms of the implications for other P2P markets is warranted. For example, one would naturally wonder whether eBay, Uber, or Prosper should regulate the professional players on their platforms in a way similar to the OHOH policy implemented by Airbnb. In a market characterized by highly homogeneous goods (or services), it is reasonable to expect that professional players with their advantages may eventually crowd out small, nonprofessional

players. In such a market, restricting professional players helps nonprofessionals to survive. However, if the products or services provided by the professional and nonprofessional suppliers are highly heterogeneous, restricting the participation of professional players does not help the nonprofessionals and eventually hurts the platform as a whole. We call for future studies to extend our understanding of professional players in other P2P markets. Second, our findings suggest that P2P platforms could not only effectively motivate nonprofessional players who are potentially suppressed by professional players, but also smooth the relationship with local legislators and defend the grassroots P2P nature. Our study informs policy discussions across platforms that have begun to experiment with various supplier governance strategies to address the welfare of market participants while still reaping the benefit from the growth of sharing economy.

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Endnotes

¹ Examples of professional players in other P2P markets are ample. On traditional P2P e-commerce sites such as eBay and Craigslist, professional businesses are in general allowed to sell products directly to consumers. In the P2P social media websites, such as Facebook Marketplace and Pinterest, institutional sellers and company brands increasingly dominate the supply side. As for P2P online labor marketplaces such as Freelancer and Upwork, professional proofreading companies, translator organizations, law firms, etc., rise to become an important source of skill providers.

² We do not claim one or the other in this research. As we better detail in the following sections, although our empirical evidence implies the dominance of the competition effect, we also find evidence suggesting that the dominance is less prominent or even muted in certain markets. In other words, the trade-off between differentiation and competition may favor the former in some markets. Therefore, the two effects are not mutually exclusive.

³ See <https://www.theverge.com/2015/12/4/9849242/airbnb-data-new-york-affordable-housing-illegal-hotels>, retrieved on November 22, 2021.

⁴ In Online Tables A19–A21, we report a robustness check that replicates our main analyses using an alternative definition of

professional hosts. Specifically, rather than restricting the property type to entire homes, we define professional hosts as those who operate multiple listings (entire homes, shared rooms, or private rooms) in a city in any month during our study period. The results are highly consistent. In addition, another robustness check considers some potentially noisy listings under our main definition of professional hosts. For example, there might be some entire homes under the same address (e.g., apartment units in the same building using street address) but labeled as professional host properties under our main definition. Despite the negligible number of such cases in our sample (detected using their precise geo-coordinates), we alleviate the concern by removing them from the sample. The results are, not surprisingly, qualitatively the same as our main findings and reported in Online Section A5.

⁵ We provide more details on the labeling of top and bottom nonprofessional hosts in Section 5. Briefly, a property is labeled a top nonprofessional host property if the price, that is, the nightly rate, is greater than the median nightly rate of all nonprofessional host properties in a market in a given month.

⁶ Consistent with the observation, we find that the reservations of top nonprofessional host properties increase after the policy, whereas the reservations of bottom nonprofessional properties do not change. These findings suggest that the guests who would have booked professional hosts' properties seem to have switched to the properties managed by top nonprofessional hosts, which further supports the competition between the two cohorts.

⁷ As an extension to the main findings, we rule out multiple alternative explanations, such as potential gaming behavior of professional hosts (in order to bypass the policy) and the introduction of a different set of hosts after the policy. In addition, we conduct an extensive set of robustness checks. Our conclusions persist in all these analyses. Online Table A1 summarizes all the analyses in this paper.

⁸ Certain hosts are exceptions to the policy restrictions. As specified on the policy page (Airbnb 2016), licensed hotels, hosts whose listings share a single address, hosts who help other hosts with their responsibilities, or those who share long-term rentals (at least 30 days) may require an exemption from the policy restriction. That said, Airbnb reviews the application on a case-by-case basis.

⁹ One may anticipate that professional hosts may bypass the policy by relisting their removed properties (e.g., through the help of friends and family members). However, we argue that first, it is costly to do so (in terms of legal and financial liabilities). Second, evidence from our data suggests that such hosts are a negligible subset of the population with only two out of 180,264 properties sampled across the policy-affected cities showing a change in the associated host ID after the policy. Finally, in a robustness check, we exclude more "suspicious" properties and get consistent estimates of the policy impacts. Online Tables A4 and A5 report the results.

¹⁰ We use the number of reservation days rather than the number of reservations (or bookings) as the measure of reservations throughout the paper. This is because Airbnb properties are made available as one unit, which is different from a hotel that has multiple units (rooms). In this case, the number of reservations (or bookings) is arguably a less accurate measure of the demand because a reservation can be booked for one or more days.

¹¹ In case one worries that our definition of professional hosts might bias the estimates of the policy impacts, we conduct a robustness check under a different definition that does not have a requirement on the listing type. We also search for the cases of multiple entire homes potentially sharing the same address (e.g., separate apartment units in the same building) and test the robustness of our findings by excluding these properties. Section 5.6.2 presents the discussions and reports the results that are largely consistent with our main findings.

¹² The distinction between the differentiation and competition effects of professional hosts (theocratized in this research as the tension) can be understood with the notion of vertical differentiation. In our framework, if the quality distance between professional properties and the nonprofessionals' is large and the professionals charge higher prices (i.e., "vertically differentiated"), the two types of properties are less likely to compete for the same set of guests. Our definition of the differentiation effect (of professional hosts) captures this possibility. The competition effect of professional hosts, in contrast, specifies the situation in which professional and nonprofessional properties are similar in quality and, hence, not so vertically differentiated. In this scenario, they compete for similar guests on the market. Although vertical differentiation is a useful framework, our focus is on the role of professional hosts in a home-sharing market (their impacts on the nonprofessionals and the platform).

¹³ We conduct a two-sample Kolmogorov–Smirnov (K-S) test, which rejects that the two distributions are identical (K-S statistic = 0.249, p -value < 0.01). Note that the nightly rates do not exceed \$500 per night as shown in Figure 1. This is because the 95th percentile of all prices is \$460, and the distribution has a long and flat tail. We set \$500 as the upper bound of the figure (not the max of actual distributions) to focus on the mass of the distributions.

¹⁴ Note that, in our framework, as argued earlier, the domination of differentiation does not exclude competition. If there exists some level of competition (between professional and nonprofessional hosts), the OHOH policy encourages some nonprofessionals to participate more. Our arguments here center on the consequences under the domination of differentiation. In this scenario, we argue that the policy does not increase the supply from nonprofessional hosts in a significant manner.

¹⁵ In addition to supply and prices, we also examine the policy effects on other market-level outcomes. Online Tables A25 and A26 report the policy effects on reservations and revenue, which, in our opinion, do not provide direct evidence of the dominant role of professional hosts and are, hence, not discussed in the main paper.

¹⁶ In the PSM process, we use the zip code-level characteristics before 2016 to maintain consistency across the policy-affected cities. In an alternative matching approach, we use the characteristics of zip codes in New York City and San Francisco before 2016 and those of Portland before 2017 (one may argue that the policy was implemented in Portland in 2017). The results using this alternative matching approach (available upon request) are qualitatively the same.

¹⁷ Online Table A2 compares the same set of characteristics for the full sample (i.e., before matching). One can tell that the differences between the affected and unaffected groups are quite significant before matching.

¹⁸ In a robustness check, we include additional matching covariates. The additional covariates are mainly Airbnb market outcomes, for example, the median nightly rate of nonprofessional host properties before the policy. Results using the alternative matched sample (much smaller in size) are qualitatively consistent with our main findings and reported in Online Appendix A5 (Tables A12–A14).

¹⁹ One may worry that the effect of the announcement may be different from the effect of the actual implementation. To alleviate such concerns, we repeat our main analysis by removing the time periods between the announcement and implementation in New York City and San Francisco (namely, April to October 2016). This way, we can "identify" the effect of implementation (to a certain degree) by removing the impact of the announcement. The results using this alternative specification (available upon request) are qualitatively the same as our main findings.

²⁰ Note that, although the sheer number of professional host accounts does not change after the policy, the fraction, defined as the ratio of the number of professional host accounts over the total

accounts, could either increase or decrease. We show later that the total number of host accounts increases after the policy (driven by the increased nonprofessional host accounts), which leads to a decreased fraction of professional host accounts.

²¹ For the number of properties, we show that the supply from professional hosts decreases after the policy, and the supply from nonprofessional hosts increases. It turns out that the two changes balance out (i.e., the increase of nonprofessional host supply compensates for the drop of professional host supply), leading to a largely unchanged total number of properties on the platform. In contrast, for the number of host accounts, we find that it stays relatively constant for professional hosts, whereas it increases for nonprofessional hosts after the policy. Therefore, we see that the increase in the total number of host accounts is significant.

²² The reason we do not study the differential policy effects on the supply of top versus bottom nonprofessional hosts is that, by our construction, each of the two cohorts consists of exactly 50% of the supply by nonprofessional hosts. Therefore, the policy effects on the supply of either top or bottom nonprofessional hosts is exactly the same as the policy effects on the supply of all nonprofessional hosts (as in Panel (I) of Table 5).

²³ As a robustness check, we construct a similarity measure between the properties of professional and nonprofessional hosts. Specifically, the similarity measure is the squared root of the mean squared differences in price between all pairs of professional and nonprofessional properties. Professional and nonprofessional properties are less similar in a market with a larger value of the similarity measure. Therefore, the moderating effects of this similarity measure should also take the opposite signs as the main policy effects. We repeat our moderation analysis and find consistent results based on this similarity measure. Because of the space constraints, the results are available upon request.

²⁴ Because of the space constraint, the results (tables and figures) in this section are reported in Online Sections A4 and A5, respectively.

²⁵ For a detailed explanation of the decimal degrees in geographic coordinates of latitude and longitude, please see http://wiki.gis.com/wiki/index.php/Decimal_degrees.

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